



RESEARCH PAPER

Nutrient deficiencies modify the ionic composition of plant tissues: a focus on cross-talk between molybdenum and other nutrients in *Brassica napus*

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Abstract:

The composition of the ionome is closely linked to a plant's nutritional status. Under certain deficiencies, cross-talk induces unavoidable accumulation of some nutrients, which upsets the balance and modifies the ionic composition of plant tissues. Rapeseed plants (*Brassica napus* L.) grown under controlled conditions were subject to individual nutrient deficiencies (N, K, P, Ca, S, Mg, Fe, Cu, Zn, Mn, Mo, or B) and analyzed by inductively high-resolution coupled plasma mass spectrometry to determine the impact of deprivation on the plant ionome. Eighteen situations of increased uptake under mineral nutrient deficiency were identified, some of which have already been described (K and Na, S and Mo, Fe, Zn and Cu). Additionally, as Mo uptake was strongly increased under S, Fe, Cu, Zn, Mn, or B deprivation, the mechanisms underlying the accumulation of Mo in these deficient plants were investigated. The results suggest that it could be the consequence of multiple metabolic disturbances, namely: (i) a direct disturbance of Mo metabolism leading to an up-regulation of Mo transporters such as MOT1, as found under Zn or Cu deficiency, which are nutrients required for synthesis of the Mo cofactor; and (ii) a disturbance of S metabolism leading to an up-regulation of root SO_4^{2-} transporters, causing an indirect increase in the uptake of Mo in S, Fe, Mn, and B deficient plants.

Key words: Interactions, ionome, molybdate transporter, molybdenum, nutrient deficiency, rapeseed.

Introduction

The plant ionome (defined as the mineral nutrient and trace element composition of an organism) is the result of the accumulation of elements from the soil solution (Salt *et al.*, 2008). In addition to oxygen, carbon, and hydrogen, 14 mineral elements, N, P, K, Ca, S, Mg, Fe, Cu, Zn, Mn, Mo, B, Cl, and Ni, are essential for plants. To ensure their growth

Abbreviations: ASY, abnormal shoot in youth; CNX, cofactor for nitrate reductase and xanthine deshydrogenase; Ct, threshold cycle; HKT1, high-affinity K^+ transporter; HVPT, *Hordeum vulgare* P transporters; IRT1, iron-regulated transporter 1; LCT1, low-affinity cation transporter; MoCo, molybdenum cofactor; MOT, molybdenum transporter; MPT, molybdopterin; NSC, non-selective cation channels; Q-PCR: quantitative polymerase chain reaction; SHST1, *Stylosanthes hamate* sulfate transporter; SULTR, sulfate transporter; ZIPs: ZRT, IRT-like proteins

and development, plants must acquire these nutrients to an appropriate content that enables physiological function without toxicity (Williams and Salt, 2009). Oilseed rape (*Brassica napus* L.) requires higher levels of N, K, P, S, Mn, and B than cereal crops for optimum yield (Grant and Bailey, 1993). As a consequence, winter oilseed rape is highly sensitive to deficiency in any one of these nutrients otherwise there is a reduction in yield (McGrath and Zhao, 1996; Jackson, 2000) and an impact on the quality of harvested products. For example, under S limitation conditions, the protein quality of seeds is affected with a reduction in the accumulation of S-rich seed storage proteins, and the lipid quality is also affected by a reduction in accumulation of C18 derivatives (D'Hooghe *et al.*, 2014). Although the uptake of macronutrients is most often considered, micronutrient deficiencies also impact yield in *B. napus*, as has been demonstrated for B, Mo, and Zn (Mei *et al.*, 2009). Conversely, an increase in the availability of B, Mo, and Zn increases the seed oil content and the oil quality (Mei *et al.*, 2009). However, each individual nutrient is important to consider separately because the plant yield is limited by the first nutrient that becomes limiting (Grant and Bailey, 1993). Application of high levels of one nutrient when another nutrient is deficient will, at best, be ineffective in increasing yield and at worst, lead to a yield reduction.

To maintain the internal ion concentrations required to satisfy plant needs, nutrient uptake is tightly regulated. Mineral elements are acquired by a complex network of root transporters that regulate import from the soil solution to the plants. Adjustment of nutrient uptake according to plant needs is hampered by limiting and fluctuating availability. To cope with this constraint, plants can modulate the root uptake. This uptake modulation can be achieved in several ways, such as root architecture modification, and by modulation of the quantity and activity of the root transporters. Numerous reviews have described the role of root uptake regulation in maintaining the homeostasis of ions such as N, P, and S (Gojon *et al.*, 2009), Fe and Cu (Grusak *et al.*, 1999; Puig *et al.*, 2007), Mn and Zn (Grusak *et al.*, 1999), or Ni, Mo, and Cl (Tejada-Jiménez *et al.*, 2009).

Some transporters are highly selective for elements, for example MOT1 (molybdenum transporter 1) or ASY (abnormal shoot in youth), which are two high-affinity transporters that specifically transport molybdate (Tejada-Jiménez *et al.*, 2007; Hibara *et al.*, 2013). Similarly, the yellow stripe 1 (YS1) transporter identified in *Hordeum vulgare* roots has high substrate specificity: only the Fe(III)–phytosiderophore complex has been shown to be transported (Murata *et al.*, 2006). The control of these specific transporters allows balance between nutrients by modulating the content of a single element without affecting the uptake of others.

Conversely, other transporters are non-specific and are able to transport several elements. For example, chloride channel (CLC) proteins function as anion channels and are able to transport both chloride and nitrate (De Angeli *et al.*, 2009). Therefore, in response to a given nutrient deficiency, plants that overexpress or up-regulate non-specific transporters with the aim of improving the uptake of the deficient nutrient may indirectly increase the uptake of other nutrients. Elements

of the ionome do not behave independently, but as a combination, interacting with each other due to similar chemical properties or biological roles (Baxter, 2015). Some of these interactions have been described extensively. This is the case for K^+ and Na^+ , which share several common transporters such as HKT1 (a high-affinity K^+ transporter), LCT1 (a low-affinity cation transporter) or NSCs (non-selective cation channels) that transport cations by passive diffusion (Schachtman and Liu, 1999). These transporters of both Na^+ and K^+ cause cross-talk between these two elements. Thus in saline soil, Na^+ will compete with K^+ for uptake; plants will preferentially take up Na^+ in place of K^+ (Schachtman and Liu, 1999; Malvi, 2011). The uptake of each of these elements is therefore highly dependent on the concentration of others in the soil solution. A strong interaction has also been reported between different divalent metals through the iron-regulated transporter IRT1, a member of the ZRT, IRT-like proteins (ZIPs) (Puig *et al.*, 2007; Pilon *et al.*, 2009). Indeed, this transporter is able to transport Fe but also other metals such as Zn and Mn. As a consequence, a low Fe level in the soil solution affects the uptake of the other ions and vice versa (Pilon *et al.*, 2009). In the same way, low amounts of Zn increase the uptake of Mn, Mo, and B (Billard *et al.*, 2015). Sub-optimal levels of Zn induce an up-regulation of expression of the high-affinity P transporters HVPT1 and HVPT2 (*Hordeum vulgare* P-transporters) in barley, which improves P uptake (Huang *et al.*, 2000).

Similar opposing relationships occur for nearly all elements. Indeed, under S-deficient conditions, high levels of Mo accumulate in tissues (Schiavon *et al.*, 2012). Molybdate and sulfate are chemical analogs (Dudev and Lim, 2004; Bittner, 2014) and therefore may compete for the binding site of the same transporters (Alhendawi *et al.*, 2005). Sulfate transporters have been described as being potentially involved in root uptake of Mo. Indeed, a high-affinity sulfate transporter previously identified from *Stylosanthes hamate*, termed SHST1, is responsible for the uptake of sulfate from the soil solution (Smith *et al.*, 1995) and is also able to transport molybdate (Fitzpatrick *et al.*, 2008).

On the other hand, nutrient symport processes have been described. With this kind of transport, an optimal supply of an element will lead to the uptake of other elements and create an overall improved physiological state. For example, an optimum supply of S has been reported to ensure an optimum uptake of Fe from the soil (Astolfi *et al.*, 2006). Similarly, increased rates of Mo in the growth medium were found to cause a significant increase in P content, while a decrease in P content occurred when Mo rates decreased (Zakikhani *et al.*, 2014).

The composition of the ionome is therefore closely linked to the nutritional status of the plant; a deficiency in one element can seriously disrupt the composition of the ionome due to positive and opposite interactions between elements. The main objectives of this work were therefore to determine the impact of nutrient deficiencies on the ionome of *B. napus*. The uptake of a range of macronutrients (N, P, K, Ca, S, and Mg) and micronutrients (Fe, Cu, Zn, Mn, Mo, B, and Na) was monitored in *B. napus* plants subjected to individual

nutrient deficiencies. Additionally, as Mo uptake was shown to be strongly modified by various element deficiencies, the mechanisms underlying the accumulation of Mo in these deficient plants were investigated.

Materials and methods

Growth conditions of plants

Seeds of *Brassica napus* var. Bohème were germinated on perlite over demineralized water for 2 d in the dark. Just after third leaf emergence, seedlings were transferred to a 10-l tank (ten seedlings per 10-l plastic tank) containing the following nutrient solution: 1.25 mM KNO₃, 1.25 mM Ca(NO₃)₂, 0.25 mM KH₂PO₄, 0.5 mM MgSO₄, 0.2 mM EDTA 2 NaFe, 10 µM H₃BO₃, 5 µM MnSO₄, 3 µM ZnSO₄, 0.7 µM CuSO₄, 0.7 µM (NH₄)₆Mo₇O₂₄, 0.1 µM CoCl₂, 0.04 µM NiCl₂, 0.1 mM SiO₂, 1.25 mM CaCl₂, and 0.25 mM KCl. The nutrient solution was buffered to pH 6.5 with 2 mM CaCO₃ and was renewed every 2 d. Plants were grown under greenhouse conditions, with a thermoperiod of 20/17 °C day/night and a photoperiod of 16 h. Natural light was supplemented with high-pressure sodium lamps (Philips, MASTER Green Power T400W, Amsterdam, Netherlands) supplying an average photosynthetically active radiation of 350 µmol photons m⁻² s⁻¹ at canopy height.

After 20 d of growth, plants were separated into thirteen sets. Plants were thereafter grown with the complete nutrient solution described above (control plants) or with a nutrient solution deprived of either N, K, P, Ca, S, Mg, Fe, Cu, Zn, Mn, Mo, or B (Supplementary data SD1 at JXB online). The composition of these nutrient solutions was chosen in order to maintain the same concentration of each nutrient and the same pH, as described in Maillard *et al.*, 2015.

Four independent replicates each consisting of three individual plants were harvested at the beginning of nutrient-deprivation (*t*₀) and after growth cessation (*t*_f). Growth cessation for each deprivation treatment was assessed by non-destructive estimation of plant weight obtained by daily measurements of total plant fresh weight of control and deficient plants. Leaves and petioles present at the beginning of nutrient-deprivation (referred to as 'old leaves' and 'old petioles', respectively) were distinguished during harvests from leaves appearing during deprivation (referred as 'young leaves' and 'young petioles', respectively). In order to distinguish old from young leaves and petioles during the later harvests, they were marked at *t*₀ using correction fluid. At each date of harvest (*t*₀ and *t*_f), whole roots were collected. A sample of each tissue was weighed and dried in an oven (60 °C) for 72 h for determination of dry weight (DW) and then kept for further analysis. The remaining fresh tissues were frozen in liquid nitrogen and stored at -80 °C for subsequent transcriptomic analyses.

Element analysis by mass spectrometry

Dry plant samples were ground to a fine powder using 0.4-mm diameter inox beads in an oscillating grinder (Mixer Mill MM400; Retsch, Haan, Germany) for further analysis. Samples of about 4 mg DW (representing about 60 to 80 µg N) of each plant organ sample were placed in tin capsules for total N and S analysis, which were determined using a continuous flow isotope mass spectrometer (Nu Instrument, Wrexham, UK) linked to a C/N/S analyzer (EA3000, Euro Vector, Milan, Italy).

K, Ca, P, Mg, Fe, Mn, Zn, Cu, Mo, Ni, and B were quantified by inductively high-resolution coupled plasma mass spectrometry (HR ICP-MS, Thermo Scientific, Element 2TM)

with prior microwave acid sample digestion (Multiwave ECO, Anton Paar, les Ulis, France) (800 µl of concentrated HNO₃, 200 µl of H₂O₂ and 1 ml of Milli-Q water for 40 mg DW). For the determination by HR ICP-MS, all the samples were spiked with two internal-standard solutions of gallium and rhodium for final concentrations of 10 and 2 µg l⁻¹, respectively, diluted to 50 ml with Milli-Q water to obtain solutions containing 2.0% (v/v) of nitric acid, then filtered at 0.45 µm using a teflon filtration system (Filtermate, Courtage Analyses Services, Mont-Saint-Aignan, France). Quantification of each element was performed using external standard calibration curves. The amount of a given element, *E*, in a tissue, *i*, at a given time, *t*, was calculated from dry weight, DW, and element concentration (%*E* as % of DW) as:

$$E = (\%E_{i,t} \times DW_{i,t}) / 100$$

Uptake of each nutrient by control or deficient plants was calculated as the difference between the plant content of each element (sum of all harvested tissues) between *t*₀ and *t*_f. Because the non-destructive estimation of plant growth cessation under deprivation at time *t*_f showed a lag time of 2–5 d, during which growth of control plants was maintained, the normalized uptake of each element (*U*_{NE}) by control plants at time *t*_f, was calculated as follows:

$$U_{NE} = (U_{AE} \times DW_D) / DW_C$$

where DW_C is DW of control plants, DW_D is DW of deprived plants, and *U*_{AE} is the uptake of element *E* achieved by control plants. This normalized uptake reflects the uptake expected by control plants when they reached the DW of nutrient-deprived plants at growth cessation, which provides a more rational way to compare uptake of each nutrient by these two sets of plants. When deprivation had no significant effect on growth (in the case of Mn or B deprivation), *U*_{NE} = *U*_{AE}.

RNA extraction, reverse transcriptase and quantitative PCR analysis

Total RNA was extracted and purified from 200 mg of fresh matter using the protocol previously described by Billard *et al.* (2014). For reverse transcription (RT), 1 µg of total RNA was converted to cDNA with an 'iScript cDNA synthesis kit' according to the manufacturer's protocol (Bio-Rad, Marne-la-Coquette, France). Quantitative polymerase chain reaction (Q-PCR) amplifications were performed using specific primers for each housekeeping gene and target genes (Table 1). For each pair of primers (Table 1), a threshold value and PCR efficiency were determined using a cDNA preparation diluted 10-fold. For all pairs of primers, PCR efficiency was around 100%. Q-PCRs were performed with 4 µl of 200× diluted cDNA, 500 nM of primers, and 1× SYBR Green PCR Master Mix (Bio-Rad, Marne-la-Coquette, France) in a real-time thermocycler (CFX96 Real Time System, Bio-Rad, Marne-la-Coquette, France). A program composed of 42 cycles of a denaturing step at 95 °C for 15 s followed by an annealing and extending step at 60 °C for 40 s was used. The specificity

of PCR amplification was examined by monitoring the presence of the single peak in the melting curves after Q-PCRs at the end of the 42 cycles. To confirm that the correct amplicon was produced from each pair of primers the Q-PCR product was sequenced (Eurofins, Ebersberg, Germany or Biofidal, France, according to the sequence examined).

The relative expression of the genes in each sample was compared with the control sample (corresponding to untreated plants at the same time of harvest) and was calculated with the delta-delta Ct (threshold cycle, $\Delta\Delta Ct$) method using the following equation: relative expression = $2^{-\Delta\Delta Ct}$, with $\Delta\Delta Ct = \Delta Ct_{\text{treated}} - \Delta Ct_{\text{control}}$ and with $\Delta Ct = Ct_{\text{target gene}} - \Delta Ct = Ct_{\text{target gene}} - Ct_{\text{housekeeping gene}}$ (for calculations, the geometric mean was considered between the Ct of the housekeeping genes), where Ct refers to the threshold cycle determined for each gene in the exponential phase of PCR amplification. Using this analysis method, the relative expression of genes in the other treatments was then compared with the control. On this basis, the relative expression of each gene in the control sample of the experiment was equal to 1 (Livak and Schmittgen, 2001).

Replications and statistical analysis

Experiments were conducted with four independent biological replicates, each corresponding to a pool of three individual

plants. Data for DW are given as the mean $\pm$ s.e. for $n=4$ while uptake data (difference of 'element *E*' plant content between t_0 and t_f , calculated randomly between two sets of four replicates) are given as the mean $\pm$ s.e. for $n=16$. Significant differences between control and deprived plants and time of harvest (t_0 versus t_f) were evaluated using the Student's test, as explained in the legend of each figure.

Results

Some mineral nutrient deprivations in B. napus reduced growth but increased the uptake of other mineral nutrients

Plant growth rate was significantly reduced by N-, K-, Ca-, P-, S-, Mg-, Fe-, Cu-, Zn-, and Mo-treatments relative to control plants (up to 70.1% reduction in dry weight in N-deprived plant) (Table 2). The biomass production was not significantly affected in Mn- and B-deprived plants. The uptake of macronutrients (N, P, K, Ca, S, and Mg) and micronutrients (Fe, Cu, Zn, Mn, Mo, B, and Na) was monitored in *B. napus* subject to individual nutrient deficiencies. Compared to control plants, the uptake of each element (Table 2) between t_0 (start of deprivation) and t_f (time of growth cessation in deprived plants) during deprivation was effectively reduced by about 100% for most nutrients. On the other hand, in Cu-, Zn- and B-depleted

Table 1. Q-PCR primer sets

Gene	Accession number	Forward	Reverse
EF1- α	DQ312264	5'-gcctggtatggtgtgacct-3'	5'-gaagttagcagcaccttg-3'
18S rRNA	X16077	5'-cggataaccgtagtaattctag-3'	5'-gtactcattccaattaccagac-3'
BnaSultr1.1	AJ416460	5'-agatattgcgacgaccag-3'	5'-gaaacgccagcaagaag-3'
BnaSultr1.2	AJ311388	5'-ggtgtagtcgctggaatggt-3'	5'-aacggagtggaggaagagcaa-3'
BnaMot1	XM_013892559	5'-ctcgccaggatttgactta-3'	5'-agatccccaacacgaacaag-3'
BnaAsy	XM_013816167.1	5'-tcacgctcatctccaaagt-3'	5'-agtcctacaacgcctcaaa-3'
BnaCnx1	XM_013826807.1	5'-acgaggaggagcagaagagaca-3'	5'-caacaaagtcttcagcaacca-3'

Table 2. Effects of N-, K-, Ca-, P-, S-, Mg-, Fe-, Cu-, Zn-, Mn-, Mo-, or B-deprivation on relative biomass production (as % of control plants) and relative uptake (as % of control plants) of the respective nutrients. Data are given as the mean $\pm$ s.e. ($n=4$ for biomass production, $n=16$ for uptake). Significant differences between control and nutrient-deprived plants: * $P<0.05$, ** $P<0.01$, and *** $P<0.001$. NS, not significant

Deprivation	Relative biomass production, as % of control plants	Biomass of control plants at harvest (g)	Relative uptake, as % of control plants	Uptake achieved by control plants	Number of days required to reach a significant growth reduction
-N	29.9 $\pm$ 1.46***	13.3 $\pm$ 0.3	-5 $\pm$ 0.98***	463.7 $\pm$ 26.73 mg	11
-K	49.6 $\pm$ 2.25**	12.8 $\pm$ 1.35	-0.6 $\pm$ 0.09***	464.5 $\pm$ 38.52 mg	15
-Ca	57.4 $\pm$ 1.57***	21.5 $\pm$ 1.11	-2.7 $\pm$ 0.32***	797.7 $\pm$ 25.34 mg	18
-P	32.5 $\pm$ 1.53***	13.3 $\pm$ 0.3	-15.1 $\pm$ 0.42***	54 $\pm$ 1.62 mg	15
-S	31.4 $\pm$ 1.02***	13.3 $\pm$ 0.3	-0.1 $\pm$ 0.52***	101.2 $\pm$ 3.39 μ g	15
-Mg	52.2 $\pm$ 2.43**	12.8 $\pm$ 1.35	2.5 $\pm$ 0.17***	35.7 $\pm$ 2.86 mg	15
-Fe	86.7 $\pm$ 1.92*	21.1 $\pm$ 0.66	-1.9 $\pm$ 0.06***	13.6 $\pm$ 0.8 mg	30
-Cu	81.7 $\pm$ 2.53*	13.3 $\pm$ 0.3	20.3 $\pm$ 0.54***	83.8 $\pm$ 4.5 μ g	30
-Zn	69.2 $\pm$ 1.79***	13.3 $\pm$ 0.3	22.1 $\pm$ 0.93***	400.1 $\pm$ 17.42 μ g	30
-Mn	93.8 $\pm$ 2.24NS	13.3 $\pm$ 0.3	-3.6 $\pm$ 0.12***	1237.1 $\pm$ 105.53 μ g	>30
-Mo	79.3 $\pm$ 2.37**	13.3 $\pm$ 0.3	-7.3 $\pm$ 0.18***	940.5 $\pm$ 31.94 μ g	>30
-B	91.0 $\pm$ 2.61NS	13.3 $\pm$ 0.3	27 $\pm$ 1.26***	529.2 $\pm$ 67.75 μ g	>30

plants, the respective uptake of Cu, Zn, and B reached only 20.3%, 22.1%, and 27.0% of the uptake achieved by control plants, indicating that traces of these nutrients were present in the nutrient solution (probably derived from demineralized water), and hence taken up by the plants.

The uptake of other mineral nutrients was affected negatively by some nutrient deprivations such as N and P (Fig. 1). On the other hand, eighteen situations were identified in which a given mineral nutrient deprivation increased the uptake of other nutrients, despite growth reduction or a growth similar to control plants (Fig. 1). For example, K deficiency induced a strong increase in Na ($846.7\% \pm 11.57$) and Cu ($226.4\% \pm 4.27$) uptake, relative to uptake in control plants, while Ca deprivation increased the uptake of K ($154.5\% \pm 1.96$) and Mg ($260.9\% \pm 3.24$). Mg-deprivation induced an increase in Zn and Mn uptake. In Fe-depleted plants, S, Cu, Zn, Mn, and Mo uptake were significantly increased relative to control plants (by 680% in the case of Cu). In Cu-, Mn- or B-depleted plants, only Mo uptake was stimulated, while Zn deprivation also stimulated P and Mn uptake. Only N, P, or Mo deprivation failed to induce an increased uptake of other mineral nutrients (Fig. 1).

Overall, Mo appears as the most reactive mineral nutrient and its uptake was significantly increased by S, Fe, Mn, Zn, Cu, and B deprivation by about 938%, 419%, 253%, 228%, 247%, and 230%, respectively, as summarized in Fig. 2A. Such increases in relative uptake were detected directly from leaf nutrient content irrespective of the growth rate (Fig. 2B). For example, the Mo content in old leaves was significantly increased by S-deprivation from $171.4 \pm 5.84 \mu\text{g g}^{-1}$ in control plants to $646.8 \pm 52.79 \mu\text{g g}^{-1}$. Similarly, Mo content was also higher in old leaves of Cu-, Zn-, Mn- and B-deprived plants than in control plants.

Effects of mineral nutrient deprivation on selected molybdate and sulfate transporters

As Mo uptake was strongly modified by various element deficiencies, the expression level of genes encoding molybdate and sulfate transporters that may be involved in the accumulation of Mo were investigated in deficient plants. The relative expression of genes encoding root molybdate transporters such as MOT1 and ASY was quantified by Q-PCR in nutrient-deprived plants (Fig. 3). The expression of *BnaMot1* relative to control plants was significantly up-regulated in S-, Cu-, and Zn-deprived plants by 1.57-, 2.32-, and 3.78-fold, respectively (Fig. 3B). Meanwhile, the expression of *BnaMot1* was not significantly changed under other nutrient deprivations (Fe, Mn, and B) that increased Mo uptake (Fig. 2A). Similarly, relative expression of *BnaAsy* was not affected significantly by S-, Fe-, Cu-, Zn-, Mn-, Mo-, or B-deprivation, relative to control plants (Fig. 3A).

As sulfate transporters (SULTRs) such as SULTR1.1 and SULTR1.2 have been previously described to be potentially involved in root uptake of Mo (Fitzpatrick *et al.*, 2008; Schiavon *et al.*, 2012; Bittner, 2014), the expression of genes encoding these transporters was also examined in the roots (Fig. 4). In Cu- and Zn-depleted plants, *BnaSultr1.1* (Fig. 4A) and *BnaSultr1.2* (Fig. 4B) gene expression in the roots remained similar to control plants. On the other hand, the level of *BnaSultr1.1* transcripts was strongly up-regulated by S-deprivation (by 219-fold compared to control plants) and to a lesser extent by Mn-, Fe-, Mo-, and B-deprivation (by 23-, 8-, 4-, and 3-fold, respectively). The gene expression of *BnaSultr1.2* encoding the second high-affinity sulfate transporter was also up-regulated by S-, Fe-, Mn-, and B-deprivation relative to control plants (Fig. 4B) by 13-, 10-,

Deprivation	Changed the uptake of (as % of control plants):											
	N	P	K	Ca	Mg	S	Zn	Cu	Mn	B	Mo	Na
–N	–	36.6 ± 2.02	28.3 ± 1.96	27 ± 1.99	37.4 ± 1.83	34.6 ± 1.78	29.0 ± 2.14	63.3 ± 8.05	14.7 ± 3.32	48.1 ± 2.89	19.6 ± 1.88	30.7 ± 1.12
–P	63.9 ± 2.34	–	31.9 ± 1.25	40.7 ± 1.33	49.6 ± 1.44	36.0 ± 1.33	46.1 ± 2.53	50.6 ± 2.78	26.1 ± 1.99	51.3 ± 2.51	23.4 ± 1.26	35.9 ± 1.74
–K	NS	122.3 ± 1.09	–	137.2 ± 2.11	141.2 ± 1.34	85.1 ± 1.02	144.0 ± 2.07	226.4 ± 4.27	137.4 ± 2.00	75.6 ± 1.12	65.7 ± 0.67	846.7 ± 11.57
–Ca	155.6 ± 1.96	118.2 ± 1.39	154.5 ± 1.96	–	260.9 ± 3.24	70.2 ± 0.83	118.1 ± 1.89	NS	61.7 ± 1.56	34.6 ± 1.11	28.1 ± 0.53	63.4 ± 1.18
–Mg	131.2 ± 1.7	131.4 ± 1.95	128.8 ± 1.37	116.1 ± 2.16	–	72.0 ± 1.04	226.1 ± 5.13	127.6 ± 2.08	188.0 ± 3.78	138.6 ± 2.5	135.6 ± 1.59	68.5 ± 1.01
–S	73.0 ± 2.36	NS	80.6 ± 3.22	131.5 ± 4.88	41.6 ± 1.64	–	NS	122.2 ± 4.42	71 ± 4.57	NS	938.0 ± 36.29	55.2 ± 1.33
–Zn	NS	157.2 ± 4.29	NS	124.3 ± 3.26	86.9 ± 1.48	NS	–	NS	186.7 ± 6.79	80.1 ± 2.88	228.4 ± 4.04	51.5 ± 0.88
–Cu	NS	NS	NS	NS	82.5 ± 2.38	NS	NS	–	NS	NS	247.2 ± 5.42	68.9 ± 2.98
–Mn	NS	112.3 ± 1.87	NS	125.3 ± 3.21	NS	113.9 ± 1.94	NS	NS	–	NS	253.1 ± 4.26	76.4 ± 1.71
–B	NS	NS	NS	132.4 ± 3.58	85.4 ± 1.86	NS	NS	NS	NS	–	230.1 ± 7.85	68.3 ± 1.40
–Mo	NS	NS	NS	NS	NS	NS	NS	NS	71.6 ± 1.77	NS	–	70.8 ± 1.40
–Fe	NS	NS	123.2 ± 1.94	114.9 ± 1.86	157.8 ± 3.29	219.5 ± 5.46	337.5 ± 31.73	680.1 ± 9.31	214.5 ± 5.57	116.8 ± 1.59	419.1 ± 11.12	126.6 ± 1.76

Fig. 1. Cross-talk between deprivation of one nutrient (–N, –K, –P, –Ca, –Mg, –S, –Zn, –Cu, –Mn, –B, –Mo, or –Fe) through stimulation (in green) or reduction (in red) of relative uptake (as % of uptake by control plants) of other nutrients (N, P, K, Ca, Mg, S, Zn, Cu, Mn, B, Mo, and Na). Data are given as the mean ± s.e. ($n=16$). Colour graduation: dark, intermediate, and light shading indicates significant differences between control and nutrient-deprived plants for $P<0.001$, $P<0.01$ and $P<0.05$, respectively. NS indicates that the difference is not significant. Highly significant effects ($P<0.001$) that correspond to at least a 150% increase in uptake are highlighted in bold.

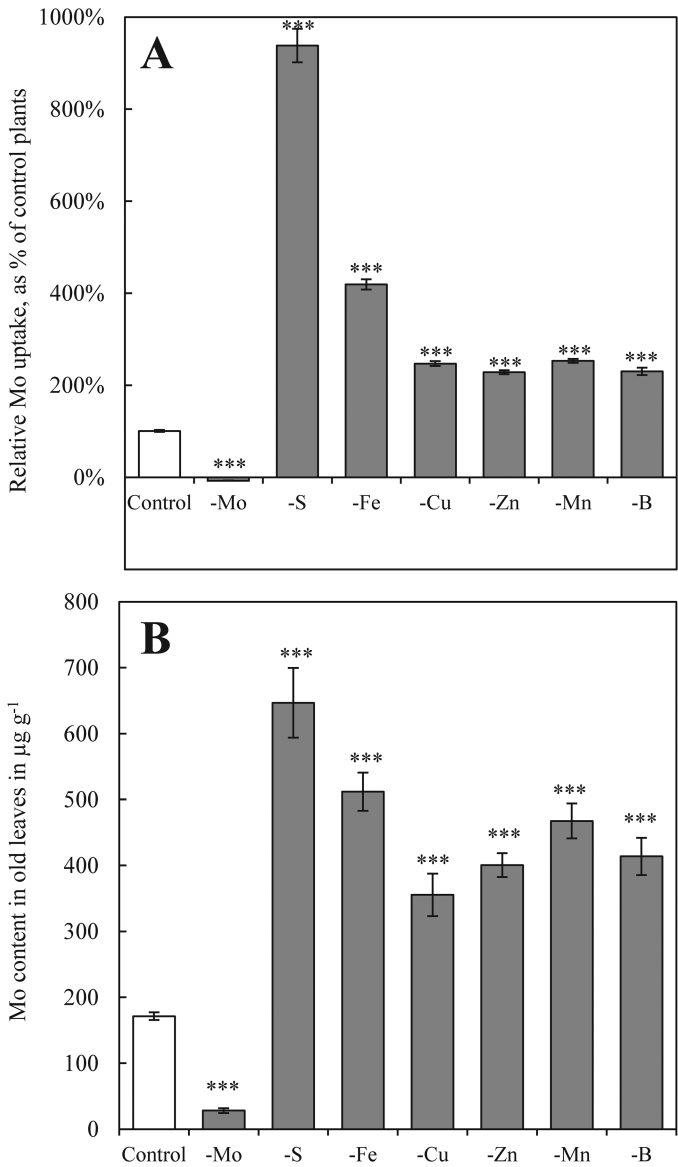


Fig. 2. (A) Relative Mo uptake (as % of control plants, for which Mo uptake was $940 \pm 32 \mu\text{g Mo plant}^{-1}$) and (B) leaf Mo content ($\mu\text{g g}^{-1}$ DW) in *B. napus* in response to Mo-, S-, Fe-, Cu-, Zn-, Mn-, or B-deprivation. Data are given as the mean $\pm$ s.e. ($n=16$ for uptake, $n=4$ for Mo concentrations). Significant differences between control and nutrient-deprived plants: * $P<0.05$, ** $P<0.01$, and *** $P<0.001$.

7- and 4-fold, respectively. As expected, the uptake of S was strongly reduced ($-100.1\% \pm 0.52$) in S-deprived plants compared to control plants (Table 2) but, more surprisingly, it was stimulated by Fe-deficiency ($+119.5\% \pm 5.46$) and to a lesser extent by Mn-deficiency ($+13.9\% \pm 1.94$) (Fig. 1). This is consistent with the enhancement of transcript accumulation of root SO_4^{2-} transporters observed in Fe- and Mn-deprived plants (Fig. 4). For B-deprived plants, the expression of *BnaSultr1.1* and *BnaSultr1.2* was also up-regulated without increasing the uptake of S. The relative S uptake in Mo-, Cu-, and Zn-deprived plants was similar to control plants.

Because Mo uptake was affected by several mineral deficiencies and considering the importance of molybdenum cofactor (MoCo) synthesis for different enzymes, as well as its complex synthesis pathway requiring several mineral nutrients, the relative expression of *BnaCnx1* (cofactor for

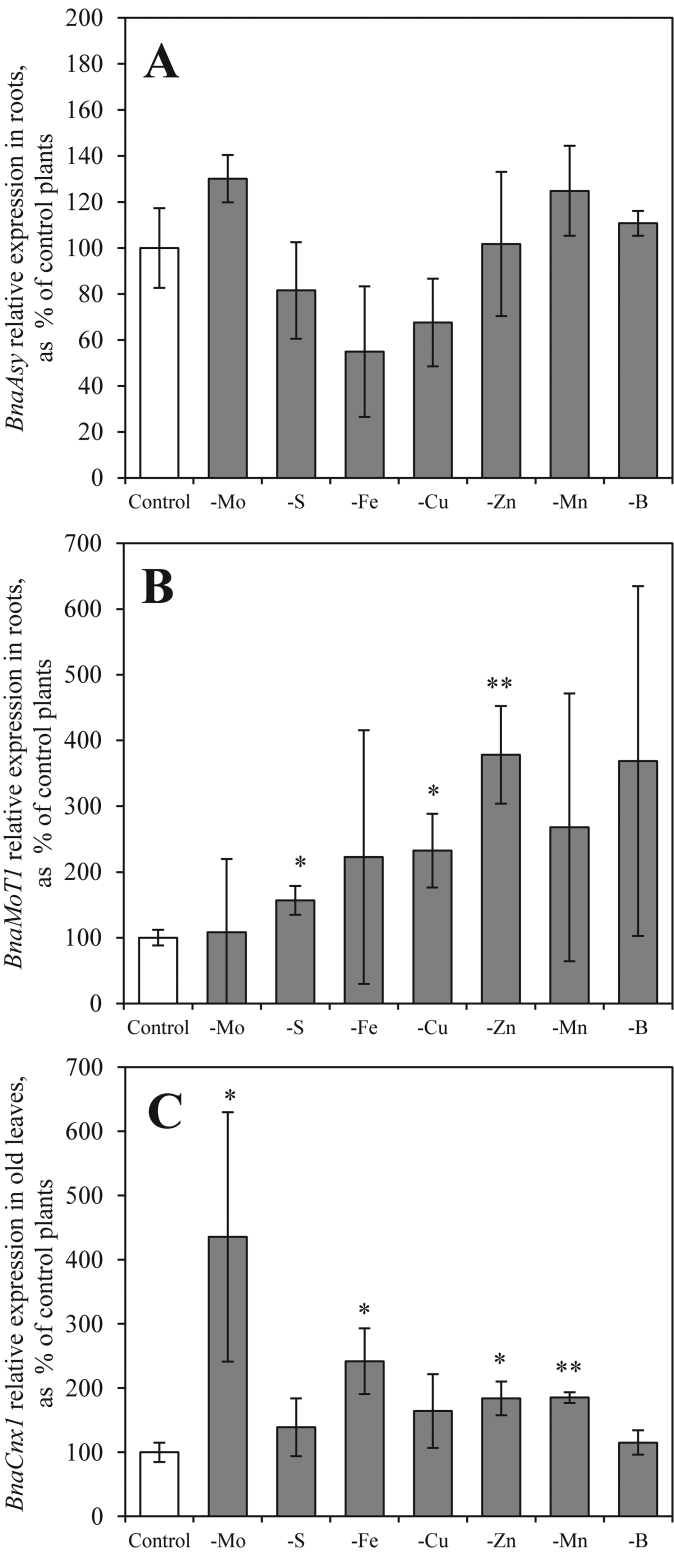


Fig. 3. Effect of Mo-, S-, Fe-, Cu-, Zn-, Mn-, and B-deprivation on relative expression of genes encoding the molybdate transporters *BnaAsy* (A) and *BnaMot1* (B) expressed in roots, and on relative expression of a key gene involved in the MoCo synthesis pathway, *BnaCnx1* (C) expressed in old leaves (data given as % of control plant expression). Data are given as the mean $\pm$ s.e. ($n=4$). Significant differences between control and nutrient-deprived plants: * $P<0.05$, ** $P<0.01$, and *** $P<0.001$.

nitrate reductase and xanthine deshydrogenase 1), a key gene involved in MoCo synthesis, was quantified. It was found to be up-regulated by Mo deficiency (4.36-fold higher than in

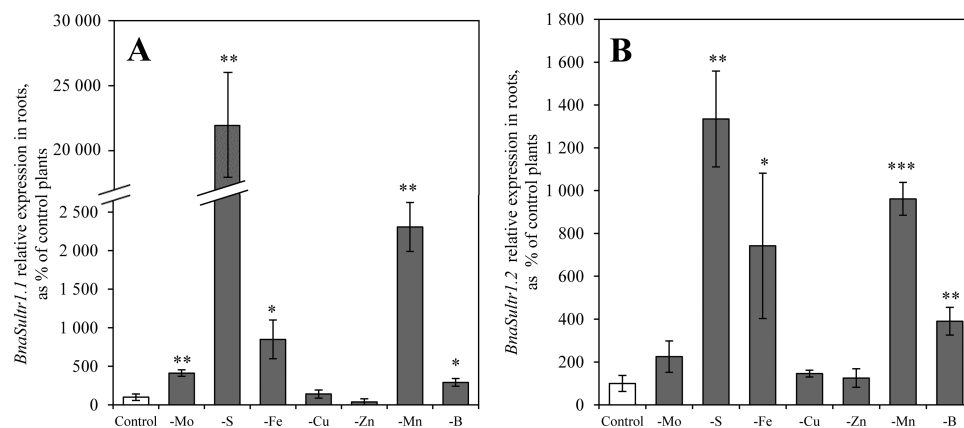


Fig. 4. Effect of Mo-, S-, Fe-, Cu-, Zn-, Mn-, or B-deprivation on relative expression of genes encoding sulfate transporters *BnaSultr1.1* (A) and *BnaSultr1.2* (B) expressed in roots of *B. napus* as % of control plants. Data are given as the mean $\pm$ s.e. ($n=4$). Significant differences between control and nutrient-deprived plants: * $P<0.05$, ** $P<0.01$ and *** $P<0.001$.

control plants) (Fig. 3C) and it also significantly increased by about 2-fold in Fe-, Zn-, and Mn-depleted plants (Fig. 3C).

Discussion

Cross-talk identified in response to individual mineral nutrient deficiencies

It is usually assumed that plant nutrient contents are tightly regulated through balanced activities of membrane transporters that mediate the uptake and distribution of nutrients in order to ensure a relative compositional homeostasis. Macronutrients such as N or P are essential for plant growth and development, being required in structure, in energy storage, and for the synthesis of proteins, which contain about 85% of the N in plant. Thus, macronutrients have a role in most plant biochemical processes. A deficiency in one of these elements has a drastic effect on growth and plant metabolism and reduces uptake of a large number of nutrients. For example, in N-deprived plants, uptake of K ($-71.7\% \pm 1.96$), S ($-65.4\% \pm 1.78$), P ($-63.4\% \pm 2.02$), and Ca ($-73.0\% \pm 1.99$) were strongly negatively affected (Fig. 1). Likewise, N, K, or S uptake was decreased under P-deficiency in *B. napus* by -36.1 , -68.1 and -64.0% , respectively (Fig. 1).

Other nutrient deficiencies increased the uptake of other elements. Indeed, cross-talk induces unavoidable accumulation of some nutrients, which upsets the balance and modifies the ionic composition of plant tissues, as previously shown with Fe or P deficiencies (Baxter et al., 2008a). Eighteen situations of significantly increased uptake under mineral nutrient deprivations were identified (Fig. 1), amongst which there was a strong accumulation of Na (of $846.7\% \pm 11.57$ relative to control plants) under K deficiency (Fig. 1). This cross-talk between K and Na is widely documented in the literature (Schachtman and Liu, 1999; Malvi, 2011) and results from up-regulation by K-deficiency of non-selective uptake systems such as HKT1, LCT1, or NSC, which transport cations by passive diffusion. Phosphate uptake, which is normally controlled by the P status of plants, was also disturbed under Zn deficiency, leading to an increase in P uptake of 157.2% in Zn-deficient plants relative to control plants (Fig. 1). This is consistent with previous work

in *Gossypium hirsutum* and in *H. vulgare* that reported a high content of P in Zn-deficient plants (Cakmak and Marschner, 1986; Huang et al., 2000). Indeed, Zn plays a role in the down-regulated expression of the HVPT1 and HVPT2 high-affinity P transporters (Huang et al., 2000). Similarly, Fe deficiency led to an accumulation of Cu, Zn, and Mn (Fig. 1). This accumulation has been reported as being caused by an up-regulation of the iron-regulated transporter, IRT1 (a member of the ZIPs divalent metal transporter family), in response to Fe-deficiency (Puig et al., 2007; Baxter et al., 2008b; Pilon et al., 2009), and which is also able to transport Zn and Mn.

Increased Mo uptake in Cu- and Zn-deficient plants: a consequence of disturbances in MoCo synthesis?

Amongst the quantified nutrients, Mo was the most affected by other deficiencies (Fig. 1). Increased Mo uptake was found under Zn- and Cu-deficiencies in *B. napus*. This is consistent with previous work reporting a higher content of Mo in *B. napus* when submitted to Cu- (Billard et al., 2014) or Zn-deficiency (Billard et al., 2015). In contrast, a Mo supply caused secondary Cu and Zn deficiencies (Ndakidemi et al., 2011). Two transporters of molybdate have been functionally characterized as being involved in Mo uptake at the soil-root interface: MOT1 in *Arabidopsis thaliana* (Tomatsu et al., 2007; Baxter et al., 2008a) and ASY in *Oryza sativa* (Hibara et al., 2013). However, to our knowledge, the functional characterization of the homologous genes coding these transporters (*BnaMot1* and *BnaAsy*) has not been reported in *B. napus*. This may limit the following interpretations, which are, however, essentially based on the strong homologies of *B. napus* genes with genes of model species (Supplementary data SD2, SD3, SD5). There was therefore no apparent link between *BnaCnx1* and *BnaMot1* gene expression, although regulation may be achieved at the post-transcriptional level. Moreover, *CNX1* (cofactor for nitrate reductase and xanthine deshydrogenase 1) gene function is mainly based on the strong homologies of *B. napus* genes with genes of *A. thaliana* (Schwarz et al., 2000; Supplementary data SD4 and SD5). The expression of *BnaAsy* remained unchanged under all tested nutrient deficiencies (Fig. 3A), which suggests that the accumulation of

Mo in response to Zn- and Cu-deficiency was not due to an up-regulation of the ASY transporter. The sequence similarity and subcellular localization of ASY suggest that it acts as a molybdate transporter, but its Mo uptake activity and Mo affinity have not been studied *in planta*. Another Mo transporter, MOT1 identified in *Clamydomonas reinhardtii* (Tejada-Jiménez *et al.*, 2007) and *A. thaliana* (Tomatsu *et al.*, 2007; Baxter *et al.*, 2008a), might be necessary for uptake of Mo from the soil. The levels of *BnaMot1* transcripts were increased in Cu- and Zn-deprived plants by 2.32-fold and 3.78-fold, respectively, compared to control plants (Fig. 3B). An up-regulation of *BnaMot1* gene expression has already been reported in Cu-depleted plants in *B. napus* (Billard *et al.*, 2015), but to our knowledge has not been previously described under Zn-deficiency. Overall, this suggests that the increase in Mo uptake in Cu- and Zn-deprived plants may at least be the result of an up-regulation of *BnaMot1* gene expression in roots.

Molybdenum is required for the synthesis of the MoCo factor found in at least six key molybdo-enzymes in plants: nitrate reductase, aldehyde oxidase, sulfite oxidase, xanthine oxidase, nitrogenase, and mitochondrial amidoxime-reducing component (mARC) (as reviewed by Schwarz and Mendel, 2006; Schwarz *et al.*, 2009; Llamas *et al.*, 2011; Bittner, 2014). The MoCo biosynthetic pathway involves at least four steps

(Fig. 5) requiring different elements, such as Cu and Zn but also S and Fe (Schwarz and Mendel, 2006; Schwarz *et al.*, 2009; Llamas *et al.*, 2011; Bittner, 2014). Although the role of Zn is unclear, it could be involved in stabilization of the CNX5 protein, which sulfurates molybdopterin (MPT) synthase for the next insertion of a dithiolene group (Llamas *et al.*, 2011). An atom of Cu is required to form the Cu–MPT complex (Fig. 5). Once transferred from CNX1G to CNX1E, the adenylation is cleaved in the presence of Mg^{2+} or Zn^{2+} . In the final step, the CNX1-E domain catalyzes the insertion of Mo into MPT in place of Cu, thus yielding physiologically active MoCo (Fig. 5). Cu could potentially act as a protecting agent for the reactive dithiolate before the insertion of Mo, as suggested by purification of an *in vivo*-formed Cu–MPT complex (Kuper *et al.*, 2004). It may also have a role in Mo insertion. Therefore, MoCo biosynthesis could be affected under Zn- or Cu-limited conditions.

The *BnaMot1* up-regulation leading to the accumulation of Mo in the case of Zn and Cu deficiencies may be explained by a defect in MoCo synthesis (Fig. 3B, Fig. 5). CNX1, the major enzyme of MoCo synthesis, has been suggested to have a role as a Mo sensor, being involved in the regulation of MOT1 activity (Tejada-Jiménez *et al.*, 2009). However, there is no experimental evidence to confirm this

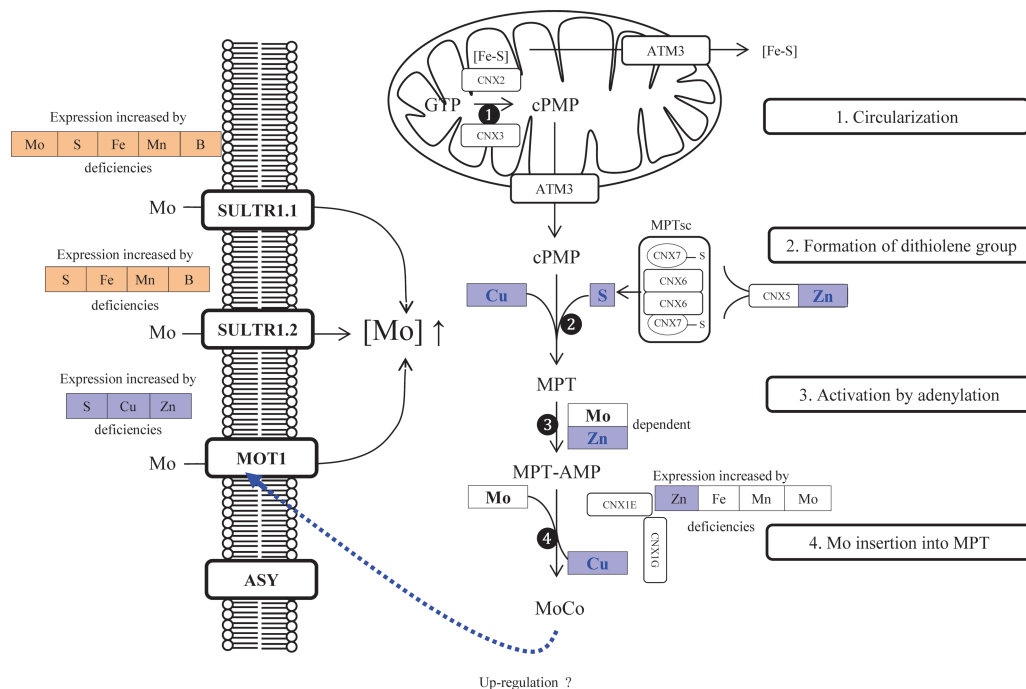


Fig. 5. Cross-talk between S, Fe, B, Mn, Cu, Zn, and Mo during MoCo synthesis. Under S-, Fe-, Mn-, and B-deficiencies (indicated in orange), the increase in Mo uptake seems to be an indirect cause of the prior up-regulation of the *BnaSutr1.1* and *BnaSutr1.2* genes, which encode root sulfate transporters. In contrast, under Cu- and Zn-deficiencies (indicated in blue), accumulation of Mo might be due to feedback signals from disturbances in MoCo synthesis (due to low Cu and Zn contents), which could induce an up-regulation of the *BnaMot1* gene (encoding the MOT1 transporter). The involvement of S, Fe, Cu, Zn, and Mo are presented sequentially in the four steps of the MoCo biosynthesis pathway. (1) The proteins CNX2 and CNX3 catalyze the conversion of GTP to cPMP. cPMP is then exported from the mitochondria by the mitochondrial ABC transporter ATM3. (2) In the cytosol, MPT-synthase composed of CNX6 and CNX7 subunits transfers two atoms of S to cPMP to form MPT. MPT synthase is then sulfurated again by the MPT synthase sulfurylase (CNX5), which contains one Zn atom per monomer. At the end of this step, an atom of Cu is bound to MPT to protect the reactive dithiolate before the insertion of Mo. (3) MPT is activated by adenylation to form MPT-AMP. This step is Mo- and Zn-dependent. (4) In the final step, MPT-AMP is transferred to the CNX1-E domain, which cleaves the adenylate from MPT and then catalyzes the insertion of Mo in place of Cu to form MoCo. Abbreviations: ATM, ABC transporters of the mitochondria; ASY, abnormal shoot in youth; CNX, cofactor for nitrate reductase and xanthine dehydrogenase; cPMP, cyclic pyranopterin monophosphate; [Fe-S], iron-sulfur cluster; GTP, guanosine triphosphate; MOT, molybdenum transporter; MPT, molybdopterin; MPT-AMP, adenylylated molybdopterin; MPTsc, molybdopterin synthase complex; SULTR, sulfate transporter.

hypothesis (Tejada-Jiménez *et al.*, 2009). *BnaCnx1* expression was up-regulated in Fe-, Zn-, Mn-, and Mo-deficient plants whereas *BnaMot1* expression was up-regulated only in the case of Zn-, Cu-, and S-deficiencies (Fig. 3C). There was therefore no apparent link between *BnaCnx1* and *BnaMot1* gene expression, although regulation may be achieved at the post-transcriptional level.

The increase in Mo uptake under Cu- and Zn-deficiencies could be the result of a direct disturbance of Mo metabolism leading to an up-regulation of the MOT1 transporter (Fig. 5).

Increased Mo uptake in S-, Fe-, Mn-, and B-deficient plants: an indirect consequence of up-regulation of root sulfate transporters

Under our conditions, Mo uptake was also significantly increased by S-, Fe-, Mn-, and B-deficiencies in *B. napus*. S deficiency strongly increased Mo uptake by 938% (Fig. 2), and this result agrees with several findings reporting that S-deficiency increases Mo uptake, as reported for several plant species (Hunashikatti *et al.*, 2000; Balik *et al.*, 2006; Schiavon *et al.*, 2012). Under our conditions, Mo uptake was significantly higher in different metal-deprived plants (Mn and Fe) (Fig. 2). Fe deficiency enhanced Mo uptake by 419% relative to control plants (Fig. 2). An inverse relationship between the shoot contents of Fe and Mo has been highlighted already by Zakikhani *et al.* (2014), with the highest Fe content in shoots being reported under a low-Mo supply and vice versa. In contrast, Baxter *et al.* (2008a) reported that the Mo content in shoot tissues was reduced in Fe-deficient plants. It was then suggested that this could be due to acidification of the rhizosphere in response to Fe deficiency, which could lead to a decrease in Mo solubility by the adsorption of molybdate by iron oxides leading to a decrease in Mo uptake (Baxter *et al.*, 2008a). However, under our conditions, the pH of the nutrient solution was maintained constant to avoid changes in the availability of nutrients. It had also been shown that Mn and Mo uptake are inversely proportional: indeed, a Mn deficiency has also been observed in response to a high Mo supply (Ndakidemi *et al.*, 2011). In agreement with these results, an increase in the Mo uptake of 253% in Mn-deprived plants was recorded under our conditions (Fig. 2). To a lesser extent, the uptake of Mo was also enhanced by B deprivation (+230%). Likewise, Ndakidemi *et al.* (2011) reported that Mo supply decreased B uptake in roots of *P. vulgaris*.

Compared to control plants, the level of *BnaAsy* transcripts remained unchanged in S-, Fe-, Mn-, and B-deprived plants (Fig. 3A), whereas *BnaMot1* expression was only increased in response to S-deficiency (Fig. 3B). Consistent with this, it has previously been reported that the level of *BnaMot1* transcripts was higher in S-deficient *Triticum aestivum* (Shinmachi *et al.*, 2010). Finally, Mo-deficiency did not induce an accumulation of *BnaMot1* transcripts (Fig. 3B), which is a similar result to that reported in *A. thaliana* (Ide *et al.*, 2011).

An alternative transport system ensuring the provision of Mo to the cell should therefore be considered. Prior to 2007, molecular mechanisms controlling Mo uptake were unknown (Hawkesford and Barraclough, 2011), so it was thought that

molybdate transport occurred non-specifically through sulfate transporters, due to the fact that molybdate is similar to sulfate in terms of net charge, structure, and hydrogen-binding properties (Dudev and Lim, 2004; Bittner, 2014). A high-affinity sulfate transporter previously identified from *S. hamate*, termed SHST1, is responsible for the uptake of sulfate from the soil solution (Smith *et al.*, 1995), and is also able to transport molybdate (Fitzpatrick *et al.*, 2008). SULTR1.1 and SULTR1.2, both homologs of SHST1, could be implicated in Mo uptake. Indeed, in response to S-, Fe-, Mn-, Mo-, and B-deprivation, gene expression of *BnaSultr1.1*, and to a lesser extent *BnaSultr1.2*, was increased in roots (Fig. 4A, B). The accumulation of Mo in S-, Fe-, Mn-, and B-deficient plants may be the indirect consequence of this increase in gene expression of sulfate transporters. Moreover, as expected, the up-regulation of the *BnaSultr1.1* gene led to an increase in S uptake in Fe- and Mn-deficient plants (Fig. 1, higher for Fe- than for Mn-deficiency), although this was not directly proportional to the level of *BnaSultr1.1* over-expression (Fig. 4, higher for Mn than for Fe), which could be the result of different post-transcriptional regulation. The interaction between S and Fe metabolisms is important, as most of the metabolically active Fe is bound to S in Fe-S clusters (Frieri *et al.*, 2013). The main function of Fe-S cluster proteins is electron transfer, with a key function in respiratory-chain complexes in mitochondria and in the photosynthetic apparatus in the chloroplasts (Frieri *et al.*, 2013). Fe-S cluster proteins are also involved in MoCo factor synthesis. S and Fe are therefore intrinsically linked, with the S availability affecting the uptake of Fe (Astolfi *et al.*, 2006). S deprivation has been reported to limit Fe-deficiency responses in *Solanum lycopersicum* (Zuchi *et al.*, 2009). Similarly, and to a lesser extent, S and Mn metabolism are linked. Manganese is also involved in redox reactions within the photosynthetic electron transport system in chloroplasts. Mn-deficiency, just like Fe-deficiency, could therefore lead to an up-regulation of root sulfate transporters. Overall, the increase in Mo uptake could result from S metabolism disturbances leading in turn to S transporter up-regulation, and indirectly causing Mo accumulation in S-, Fe-, Mn-, and B-deprived plants (Fig. 5).

Conclusions

The use of individual mineral nutrient deficiencies provides a way to decipher the multiple interactions that may exist between the uptake and metabolic use of minerals. Eighteen different interactions were identified at the uptake level, some of which have already been described previously (K and Na, S and Mo, Fe, Zn and Cu). The uptake of Mo was the most frequently affected, being increased under S-, Fe-, Cu-, Zn-, Mn-, or B-deprivation, and the results suggest that this could be the consequence of different metabolic disturbances: (i) a direct disturbance of Mo metabolism leading to up-regulation of Mo transporters such as MOT1, as found under Zn- or Cu-deficiency, which are nutrients required for MoCo synthesis; and (ii) a disturbance of S metabolism leading to up-regulation of root SO_4^{2-} transporters, indirectly causing an increased uptake of Mo in S-, Fe-, Mn-, and B-deficient plants.

Supplementary data

Supplementary data are available at *JXB* online.

SD1. Compositions of the different nutrient solutions, derived from Hoagland nutrient solution, used to expose *Brassica napus* to deprivation of individual nutrients..

SD2. Multiple alignment of ASY amino acid sequences from *Arabidopsis thaliana*, *Oryza sativa*, and *Brassica napus*.

SD4. Multiple alignment of CNX1 amino acid sequences from *Arabidopsis thaliana* and *Brassica napus*.

SD5. Unrooted phylogenetic tree of amino acid sequences of Mo-associated proteins (ASY, MOT1 and CNX1) from *Arabidopsis thaliana*, *Oryza sativa*, and *Brassica napus*.

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